

Merging Dynamic Covariates: Predicting density and assessing factors driving marbled murrelet distribution in Puget Sound using hierarchical distance sampling

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Required Packages

```
library(ggplot2) # plotting
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
library(sf) # spatial
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.6.0
```

```
## v lubridate  1.9.4      v tibble     3.3.0
```

```
## v purrr      1.1.0      v tidyr      1.3.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Required Files Previously Made or Downloaded

- grid2km_hex.gpkg
- lat_long_times.rds
- grid_tides.rds (made in “04-TidalData”)
- track_tides.rds (made in “04-TidalData”)
- Dynamicgrid_data.RData (made in “03-DynamicCovariates”)

```
hex_grids <- st_read("GIS_COVS/grid2km_hex.gpkg")
```

```
tracklines <- st_read("../GIS_COVS/all_tracklines_2026_final.gpkg")
```

```
lat longs <- readRDS("../GIS_COVS/lat_long_times.rds")
```

```
tidal_grids <- readRDS("GIS_COVS/grid_tides.rds")
```

```
tidal_tracks <- readRDS("GIS_COVS/track_tides.rds")
```

```
load("GIS_COVS/Dynamicgrid_data.RData")
```

Tidal

```
monyears <- unique(lat_long$monyear)
length(monyears)
```

```
## [1] 100
```

```
tidal_gclean <- tidal_grids %>%
  mutate(monyear = toupper(monyear)) %>%
  filter(monyear %in% monyears)

length(unique(tidal_gclean$monyear))
```

```
## [1] 100
```

```
length(unique(tidal_gclean$ID))
```

```
## [1] 1545
```

```
tidal_tclean <- tidal_tracks %>%
  mutate(monyear = toupper(monyear)) %>%
  filter(monyear %in% monyears)

length(unique(tidal_tclean$monyear))
```

```
## [1] 100
```

```
length(unique(tidal_tclean$trip_date))
```

```
## [1] 8095
```

Grid Chem

```
chem_clean <- hex_chem %>%
  st_drop_geometry() %>%
  dplyr::select(-area) %>%
  pivot_longer(
    cols = matches("(do|dic)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "(do|dic).([A-Z]{3}).(\\d{4})" %>%
  mutate(monyear = paste0(mon, "-", year)) %>%
  filter(source == "do" & monyear %in% monyears) %>%
  dplyr::select(-source) %>%
  rename(do = value)

length(unique(chem_clean$monyear))
```

```
## [1] 100
```

```
length(unique(chem_clean$ID))
```

```
## [1] 1545
```

Track Chem

```
chem_clean_track <- track_chem %>%
  st_drop_geometry() %>%
  dplyr::select(-total_length) %>% # Remove poly_overlapkm2
  pivot_longer(
    cols = matches("(do|dic)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "(do|dic).([A-Z]{3}).(\\d{4})" %>%
  mutate(moneyear = paste0(mon, "-", year),
         moneyear2 = toupper(format(as.Date(date), "%b-%Y"))) %>%
  filter(source == "do" & moneyear == moneyear2) %>%
  dplyr::select(-c(source, moneyear2)) %>%
  rename(do = value) %>%
  group_by(trip_date, date, mon, year, moneyear) %>%
  summarise(do = mean(do), .groups="drop") %>%
  ungroup()

length(unique(chem_clean_track$moneyear))
```

```
## [1] 100
```

```
length(unique(chem_clean_track$trip_date))
```

```
## [1] 8095
```

Grid Phy

```
phy_clean <- hex_phy %>%
  st_drop_geometry() %>%
  dplyr::select(-area) %>% # Remove poly_overlapkm2
  pivot_longer(
    cols = matches("(ss|st|sig)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "(ss|st|sig).([A-Z]{3}).(\\d{4})" %>%
  mutate(moneyear = paste0(mon, "-", year)) %>%
  filter(moneyear %in% moneyears) %>%
  pivot_wider(names_from = source,
              values_from = value)

length(unique(phy_clean$moneyear))
```

```
## [1] 100
```

```
length(unique(phy_clean$ID))
```

```
## [1] 1545
```

Track Phy

```
phy_clean_track <- track_phy %>%  
  st_drop_geometry() %>%  
  dplyr::select(-total_length) %>% # Remove poly_overlapkm2  
  pivot_longer(  
    cols = matches("^(ss|st|sig)."),  
    names_to = c("source", "mon", "year"),  
    names_pattern = "(ss|st|sig).([A-Z]{3}).(\\d{4})" %>%  
    mutate(monyear = paste0(mon, "-", year),  
           monyear2 = toupper(format(as.Date(date), "%b-%Y"))) %>%  
    filter(monyear == monyear2) %>%  
    pivot_wider(names_from = source,  
               values_from = value) %>%  
    select(-monyear2) %>%  
    group_by(trip_date, date, mon, year, monyear) %>%  
    summarise(st = mean(st), ss = mean(ss), .groups="drop") %>%  
    ungroup()
```

```
length(unique(phy_clean_track$monyear))
```

```
## [1] 100
```

```
length(unique(phy_clean_track$trip_date))
```

```
## [1] 8095
```

```
bio_clean <- hex_bio %>%  
  st_drop_geometry() %>%  
  dplyr::select(-area) %>%  
  pivot_longer(  
    cols = matches("(z1|z2|dia)."),  
    names_to = c("source", "mon", "year"),  
    names_pattern = "(z1|z2|dia).([A-Z]{3}).(\\d{4})" %>%  
    mutate(monyear = paste0(mon, "-", year)) %>%  
    filter(monyear %in% monyears) %>%  
    pivot_wider(names_from = source,  
               values_from = value)
```

```
length(unique(bio_clean$monyear))
```

Grid Bio

```
## [1] 100
```

```
length(unique(bio_clean$ID))
```

```
## [1] 1545
```

```
bio_clean_track <- track_bio %>%
  st_drop_geometry() %>%
  dplyr::select(-total_length) %>%
  pivot_longer(
    cols = matches("(z1|z2|dia)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "(z1|z2|dia).([A-Z]{3}).(\\d{4})" %>%
    mutate(moneyear = paste0(mon,"-",year),
           moneyear2 = toupper(format(as.Date(date), "%b-%Y"))) %>%
    filter(moneyear == moneyear2) %>%
    pivot_wider(names_from = source,
               values_from = value) %>%
    select(-moneyear2) %>%
    group_by(trip_date, date, mon, year, moneyear) %>%
    summarise(dia = mean(dia), z1 = mean(z1), z2 = mean(z2), .groups="drop") %>%
    ungroup()
```

```
length(unique(bio_clean_track$moneyear))
```

Track Bio

```
## [1] 100
```

```
length(unique(bio_clean_track$trip_date))
```

```
## [1] 8095
```

```
turb_clean <- hex_par %>%
  st_drop_geometry() %>%
  dplyr::select(-area) %>%
  pivot_longer(
    cols = matches("(turb|par)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "(turb|par).([A-Z]{3}).(\\d{4})" %>%
    mutate(moneyear = paste0(mon,"-",year)) %>%
    filter(moneyear %in% moneyears) %>%
    pivot_wider(names_from = source,
               values_from = value)
```

```
length(unique(turb_clean$moneyear))
```

Grid PAR

```
## [1] 100
```

```
length(unique(turb_clean$ID))
```

```
## [1] 1545
```

```
turb_clean_track <- track_par %>%
  st_drop_geometry() %>%
  dplyr::select(-total_length) %>%
  pivot_longer(
    cols = matches("(turb|par)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "(turb|par).([A-Z]{3}).(\\d{4})" %>%
    mutate(monyear = paste0(mon, "-", year),
           monyear2 = toupper(format(as.Date(date), "%b-%Y"))) %>%
  filter(monyear == monyear2) %>%
  pivot_wider(names_from = source,
             values_from = value) %>%
  select(-monyear2) %>%
  group_by(trip_date, date, mon, year, monyear) %>%
  summarise(par = mean(par), .groups="drop") %>%
  ungroup()

length(unique(turb_clean_track$monyear))
```

Track PAR

```
## [1] 100
```

```
length(unique(turb_clean_track$trip_date))
```

```
## [1] 8095
```

Combining all data

```
dynamic_data_grid <- bio_clean %>%
  full_join(turb_clean, by = c("ID", "numid", "monyear", "year", "mon")) %>%
  full_join(phy_clean, by = c("ID", "numid", "monyear", "year", "mon")) %>%
  full_join(chem_clean, by = c("ID", "numid", "monyear", "year", "mon")) %>%
  full_join(tidal_gclean, by = c("ID", "monyear")) %>%
  full_join(hex_suit %>%
    dplyr::select(-c(layer, area)) %>%
    st_drop_geometry(),
    by = c("ID", "numid", "year"))
```

```
dynamic_data_track <- bio_clean_track %>%
  full_join(turb_clean_track, by = c("trip_date", "date",
                                     "monyear", "year", "mon")) %>%
  full_join(phy_clean_track, by = c("trip_date", "date",
                                     "monyear", "year", "mon")) %>%
  full_join(chem_clean_track, by = c("trip_date", "date",
                                     "monyear", "year", "mon")) %>%
  full_join(tidal_tclean, by = c("trip_date", "monyear")) %>%
  full_join(track_suit %>%
    dplyr::select(-c(total_area)) %>%
    st_drop_geometry(),
    by = c("trip_date", "date", "year"))
```

Save

```
saveRDS(dynamic_data_grid, "dynamic_data_clean_grids.rds")
saveRDS(dynamic_data_track, "dynamic_data_clean_tracks.rds")
```